

SUMMARY

I enjoy figuring out how things work. I study how biological and artificial neural networks represent information, learn, and produce behavior. I'm currently building multimodal neural foundation models as a founding machine learning researcher at Metamorphic. My research connects computational and systems neuroscience, generative modeling, mechanistic interpretability, and robotics. I enjoy working with people across areas of expertise to test theories and build the tools and infra these questions require.

EDUCATION

Ph. D. Computational & Systems Neuroscience | University College London | London, UK | 2025
B.A. Neuroscience | Boston University | Boston, MA, USA | 2015

PROFESSIONAL EXPERIENCE

Founding Machine Learning Researcher, Enigma / Metamorphic Palo Alto, CA, USA	2025/09 – Present
Bogue Fellow Researcher, UCL - Anthropic San Francisco, CA, USA	2025/01 – 2025/03
Data Scientist, Sainsbury Wellcome Centre University College London, London, UK	2020/11 – 2021/08
Software Developer, CortexLab & International Brain Lab University College London, London, UK	2018/10 – 2020/08
Technical Associate I/II, Wilson Lab Massachusetts Institute of Technology, Cambridge, MA, USA	2016/06 – 2018/06
Research Assistant, Perceptual Neuroimaging Lab Boston University, Boston, MA, USA	2013/06 – 2014/06
Research Assistant, Calakos Lab Duke University, Durham, NC, USA	2012/05 – 2012/08

PUBLICATIONS

Conference & Workshop Papers

F. D'Agostino*, A. Gilbert*, S. Keller*, ... **J. Bhagat**, et al. [Evaluating Neural Data Tokenizers: A Framework for Assessing Learned Representations of Spiking Activity](#). *NeurIPS 2026 (Oral)*.
J. Bhagat*, S. Molas-Medina*, G. Giglemiani, S. Heimersheim. [Compressed computation is not computation in superposition](#). *NeurIPS Mechanistic Interpretability Workshop 2025*.
J. Bhagat, A. G. Pouget, S. Molas-Medina. [A pipeline for interpretable neural latent discovery](#). *NeurIPS Data on the Brain & Mind Workshop 2025*.

Manuscripts Under Review

F. D'Agostino, **J. Bhagat**, et al. SpikeDiT: Generative neural system identification at spike-train resolution. (*In Review*).
A. G. Pouget*, **J. Bhagat***, et al. Reinforcement learning models of ethological mouse behavior reveals static and dynamic social preferences and policies. (*In Review*).

Journal Articles & Preprints

D. Campagner*, [J. Bhagat*](#), G. Lopes*, et al. [Aeon: An open-source platform to study the neural basis of ethological behaviors over naturalistic timescales](#). *Biorxiv* 2025 & *In Press*.

International Brain Laboratory, K. Banga, J. Benson, [J. Bhagat](#), et al. [Reproducibility of in-vivo electrophysiological measurements in mice](#). *eLife* 2025.

N. A. Steinmetz*, C. Aydin*, A. Lebedeva*, M. Okun*, M. Pachitariu*, [J. Bhagat](#), et al. [Neuropixels 2.0: A high-density probe for stable, long-term brain recordings](#). *Science* 2021.

[J. Bhagat*](#), M. J. Wells*, K. D. Harris, M. Carandini, C. P. Burgess. [Rigbox: An open-source toolbox for probing neurons and behavior](#). *eNeuro* 2020.

Conference Abstracts

D. Campagner*, [J. Bhagat*](#), G. Lopes*, et al. [Aeon: An open-source platform for testing normative models of natural behaviors and their neural implementations](#). *Cosyne* 2025.

[J. Bhagat*](#), M. J. Wells*, K. D. Harris, M. Carandini, C. P. Burgess. [Rigbox: An open-source toolbox for probing neurons and behavior](#). *SFN* 2019.

[J. Bhagat](#), H. Penagos, M. Wilson. [LSTM neural networks for LFP event detection and classification in the rodent hippocampal-cortical network](#). *MIT BCS Symposium* 2018.

[J. Bhagat](#), C. Varela, F. Flores, J. Zhang, M. Wilson. [Machine Learning Techniques to Improve Analyses of Neural Spike Data](#). *MIT Intelligence Quest* 2018. [[Press Release](#)]

OPEN-SOURCE PROJECTS

[Neuronauts](#): An educational outreach program for teaching teenagers fundamentals in engineering, computer science, neuroscience, and artificial intelligence. (*Founder*)

[nanoGPT](#): A minimal (nanomal?) repository containing code for building, training, and running nanoGPT. (Sole creator, developer, and maintainer)

[Wall-E-GPT](#): Python and Arduino code for a GPT-controlled, semi-autonomous rover robot running on a Raspberry Pi. (Sole creator, developer, and maintainer)

[aeon_mecha](#): [Project Aeon](#)'s main Python library for interfacing with acquired experiment data. (Creator, developer, maintainer: active)

[aeon.experiments](#): [Project Aeon](#)'s main Bonsai and C# library for running behavioral neuroscience experiment workflows. (*Developer, maintainer: active*)

[ibllib](#): The International Brain Laboratory's core shared Python libraries for data management and neural electrophysiology acquisition $\leftrightarrow$ analysis pipeline. (Developer, maintainer)

[MatchMentor](#): Football video analysis AI to democratize player training. (Creator, developer, and maintainer: active)

[Rigbox](#): A MATLAB and C based toolbox for running behavioral neuroscience experiments and managing data. (Developer, maintainer)

[Burgess Steering Wheel Setup](#). Hardware designs and assembly instructions for a rig used to run the [Burgess Steering Wheel Task](#). (Contributor)

[J_Clust](#). A MATLAB complete spike sorting package. (Sole creator, developer, and maintainer)

TALKS & TUTORIALS

[Neuroverse Panel Discussion: "Can Computational Models & Machine Learning Capture the Full Complexity of the Brain?"](#) Sainsbury Wellcome Centre, University College London. London, UK. 2024-10-22.

[Video-based Analysis of Animal Behavior.](#) Sainsbury Wellcome Centre, University College London. London, UK. 2023-11-29.

[Collaborative Coding and Software Development Best Practices.](#) Sainsbury Wellcome Centre, University College London. London, UK. 2023-10-25.

[Introduction to Software Development in Python.](#) Sainsbury Wellcome Centre, University College London. London, UK. 2023-09-21.

[Neuroverse Podcast: Challenges with Collaborative Neuroscience & the Impact of Generative AI.](#) London, UK. 2023-06-29.

[Introduction to High-performance Computing with Linux.](#) Sainsbury Wellcome Centre, University College London. London, UK. 2023-04-25.

International Brain Laboratory Data Organization and Architecture. Jerome L. Green Science Center, Columbia University. New York, NY. 2019-09-02.

Optimizing Tetrode Spike Sorting and Using LSTMs for LFP Event Detection. Cruciform Building, UCL. London, UK. 2018-04-06.

Git and Github for Code Versioning and Collaboration in Neuroscience. Picower Institute for Learning and Memory, MIT. Cambridge, MA. 2018-03-28.

[An Introduction to Spike Sorting.](#) Center for Brains, Minds and Machines, MIT. Cambridge, MA. 2017-03-22.

TEACHING

[Neuronauts Summer Camp – 2023.](#) Instructor. Sainsbury Wellcome Centre, University College London. London, UK. 2023-07-31 – 2023-08-04.

[Neuronauts School Club – 2022-2023.](#) Instructor. Sainsbury Wellcome Centre, University College London. London, UK. 2022 - 2023.

[SWC Bonsai Workshop – 2022.](#) Teaching Assistant. Sainsbury Wellcome Centre, University College London. London, UK. 2022-11-08 – 2022-11-11.

[SWC Ph. D. Bootcamp – 2022.](#) Teaching Assistant. Sainsbury Wellcome Centre, University College London. London, UK. 2022-09-13 – 2022-09-30.

[Neuronauts Summer Camp - 2022.](#) Head Instructor. Sainsbury Wellcome Centre, University College London. London, UK. 2022-07-25 – 2022-08-05.

[International Brain Lab Code Camp – 2019.](#) Workshop Organizer and Instructor. Jerome L. Green Science Center, Columbia University. New York, NY. 2019-09-02 – 2019-09-06.

[Neuropixels Course at UCL – 2019.](#) Workshop Teaching Assistant. Sainsbury Wellcome Centre, UCL. London, UK. 2019-04-29 – 2019-05-01.

[Neuroinformatics – Spring 2019.](#) Course Teaching Assistant. Sainsbury Wellcome Centre, UCL. London, UK. 2019.

[9.123: Neurotechnology in Action – Spring 2018](#). Guest Lecturer. Department of Brain and Cognitive Sciences, MIT. Cambridge, MA. 2018.

[9.17: Systems Neuroscience – Fall 2017](#). Guest Lecturer. Department of Brain and Cognitive Sciences, MIT. Cambridge, MA. 2017.

AWARDS

[Bogue Fellowship](#) 2024: £4000

[Fondation JFMLCT Education Outreach Fund](#) 2023: £5000

[UCL AWPO Bespoke Project Fund](#) 2022: £15000

[SWC Public Engagement Fund](#) 2022: £1000

[SWC Ph. D. Scholarship](#) 2021: £150000

PROFESSIONAL GROUPS & ACTIVITIES

[UCL NeuroAI](#). Organizing committee member. 2023-Present.

[Neuronauts](#). Founder, President. 2021-Present.

[Project Aeon](#). Head of Data Architecture. 2021-Present.

[International Brain Lab](#). Member. 2019-Present.

[Society for Neuroscience](#). Member. 2019-Present.

[Cognitive Rhythms Collaborative](#). Member. 2017-2018.

[MIT Ephys Analysis and Spike Sorting Discussion Group](#). Founder, President. 2016-2018.

[The Nerve Magazine](#). Writer and Editor. 2013-2014.

ADDITIONAL TRAINING

2025 | [MARS Scholar - AI Alignment](#) | Cambridge AI Safety Hub | Cambridge, UK

2024 | [ARENA Scholar - AI Alignment](#) | ARENA Education

2023/10 | [Cajal Ephys Acquisition Course](#) | Open Ephys / Cajal Advanced Neuroscience Training

2023 | [Machine Learning Summer School](#) | Stellenbosch University | Gordon's Bay, South Africa

2018 | [Advanced Study Program - MIT](#) | Massachusetts Institute of Technology | Cambridge, MA, USA

2016 | [Programming Languages Specialization](#) | Coursera

2016 | [Machine Learning Specialization](#) | Coursera

TECHNICAL SKILLS & KNOWLEDGE

(items in each subsection in descending, rightward rank by proficiency)

Data Engineering

ETL Pipelines

Data Lakes (Petabyte multimodal object stores)

Containerization & Workflow Orchestration (via Docker, K8s, Ray, Prefect)

Computer Science & Programming

General Programming Languages (Active Projects)

Python (including Jax, PyTorch, TensorFlow, Keras, Polars, Pandas and Scikit-learn packages)

Bonsai Rust CUDA Mojo

General Programming Languages (Past Projects)

MATLAB (including NI-DAQ mx, Data Acquisition, Signal Processing, Wavelet, Statistics and Machine Learning, and Neural Network toolboxes)

C Racket Lisp ML Ruby Javascript Java R Wolfram Julia

Database & Command Languages

SQL / MySQL Git Zsh Bash PS

Markup Languages

Markdown $\LaTeX$ ϵ HTML CSS

Data Serialization Languages

YAML JSON XML

Programming Paradigms

Procedural Object-Oriented Functional Reactive

Development Practices and Processes

Versioning (DVCS) Test-driven development (TDD) Continuous integration (CI) Continuous delivery (CD) Software Development Lifecycle (SDLC) Models: Waterfall, V, Agile (Kanban, Scrum)

Statistics & Mathematics

Data Visualization & Analysis

Dimensionality Reduction Procedures: PCA, Wavelet Analysis, t-SNE

Inferential Statistics: Frequentist & Bayesian hypothesis testing

ML & AI

Supervised Learning

Regression Methods: Logistic, Multilinear (incl. L1 / L2 norm), GLMs

Classification Methods: SVMs, Random Forests

Unsupervised Learning

Clustering Techniques: k-means, FCM, MoG, HDBSCAN, OPTICS

Deep & Reinforcement Learning

ANN Architectures: Multimodal Transformers, VAE, GAN, LSTM, RNN, CNN

Training Optimizers: Adam and variants, RMSprop, SGD, BFGS

Regularization Techniques: K-Fold and Cross-validation, Layer and Gradient Normalization, Dropout

RL Models: PPO, IQL, other Policy Gradient Methods, DQN, Distributional, TD, Bandit

Mechanistic Interpretability

Retrieval-Augmented Generation (RAG)

Distributed Training (incl. DDP, FSDP over a full 72-GPU cluster via Dstack, Skypilot)

Cloud Services (incl. Wandb, AWS (S3, EC2, ECS), GCP, HF)

Signal Processing

Basic Digital Filter Design: IIR and FIR filters.

Mechatronics

Creating basic mechatronic circuits

Designing simple PID and Kalman Filter Control Systems

Assembling experiment rigs, via 3d printing and laser-cutting

Micro-controllers/computers: Designing and building hobbyist robots using Raspberry Pis (with ROS / RosPy), LattePandas, Jetson Nanos, Arduinos, and FPGAs

Assembling computer acquisition systems from components

Wiring tetrodes and building tetrode drives

Wet Lab

Behavioral assessments on humans and rodents

Genotyping (via PCR and gel electrophoresis) on rodents

Electrophysiology and fiberphotometry neural recordings on rodents

Optogenetic and pharmacologic neural manipulations on rodents

Stereotaxic surgeries on rodents

Perfusions, histology, and post-mortem fluorescence microscopy on rodents